Experiment No: 5

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Experiment:

Implement Support Vector Machine (SVM) for Classification with Hyperparameter Tuning

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1. Dataset Source

Dataset Used: Iris Dataset

Type: Classification

Source:
https://scikit-learn.org/stable/datasets/toy_dataset.html#iris-dataset

Access Method:
```
from sklearn.datasets import load_iris
```

Justification:

- Standard real-world benchmark dataset
- Suitable for linear and non-linear classification
- Works well for evaluating SVM performance

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2. Dataset Description

2.1 Iris Dataset

Objective:
To classify iris flowers into different species using the Support Vector Machine algorithm.

Number of Instances: 150
Number of Features: 4

Features:

- Sepal length
- Sepal width
- Petal length
- Petal width

Target Variable:

Species:

- Setosa
- Versicolor
- Virginica

Characteristics:

- Multiclass classification dataset
- Numerical feature values
- No missing values
- Suitable for margin-based classifiers

Column Name	Data Type	Description
ID	Numerical (Integer)	Unique identifier for each client.
LIMIT_BAL	Numerical (Integer)	Credit limit assigned to the client (in New Taiwan Dollars).
SEX	Categorical (Integer)	Gender (1: Male, 2: Female).
EDUCATION	Categorical (Integer)	Education level (1: Graduate school, 2: University, 3: High school, 4: Others).
MARRIAGE	Categorical (Integer)	Marital status (1: Married, 2: Single, 3: Others).

3. Mathematical Formulation of the Algorithm

3.1 Support Vector Machine (SVM)

SVM finds the optimal hyperplane that maximises the margin between different classes.

Equation of a Hyperplane:

$$w \cdot x + b = 0$$

Margin:

$$Margin = \frac{2}{||w||}$$

Optimisation Objective:

$$\min \frac{1}{2}||w||^2$$

Subject to:

$$y_i(w \cdot x_i + b) \geq 1$$

3.2 Kernel Trick

Used for non-linear classification.

Common kernels:

- Linear
- Polynomial
- Radial Basis Function (RBF)

Kernel function:

$$K(x_i, x_j) = \phi(x_i) \cdot \phi(x_j)$$

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4. Algorithm Limitations

- Computationally expensive for large datasets
- Sensitive to the choice of kernel and hyperparameters
- Requires feature scaling
- Less effective when classes overlap significantly

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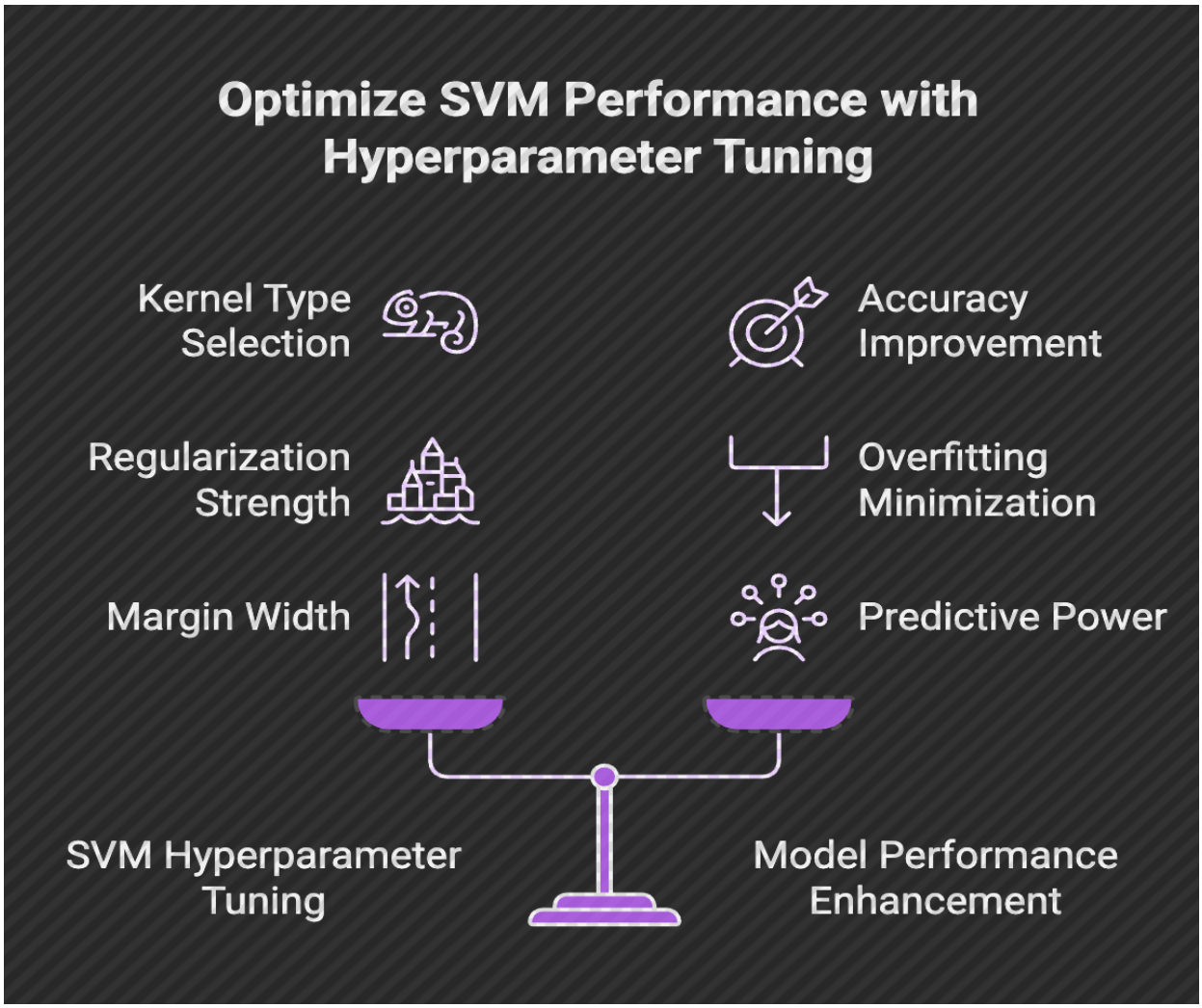
5. Methodology / Workflow

Step-by-Step Workflow:

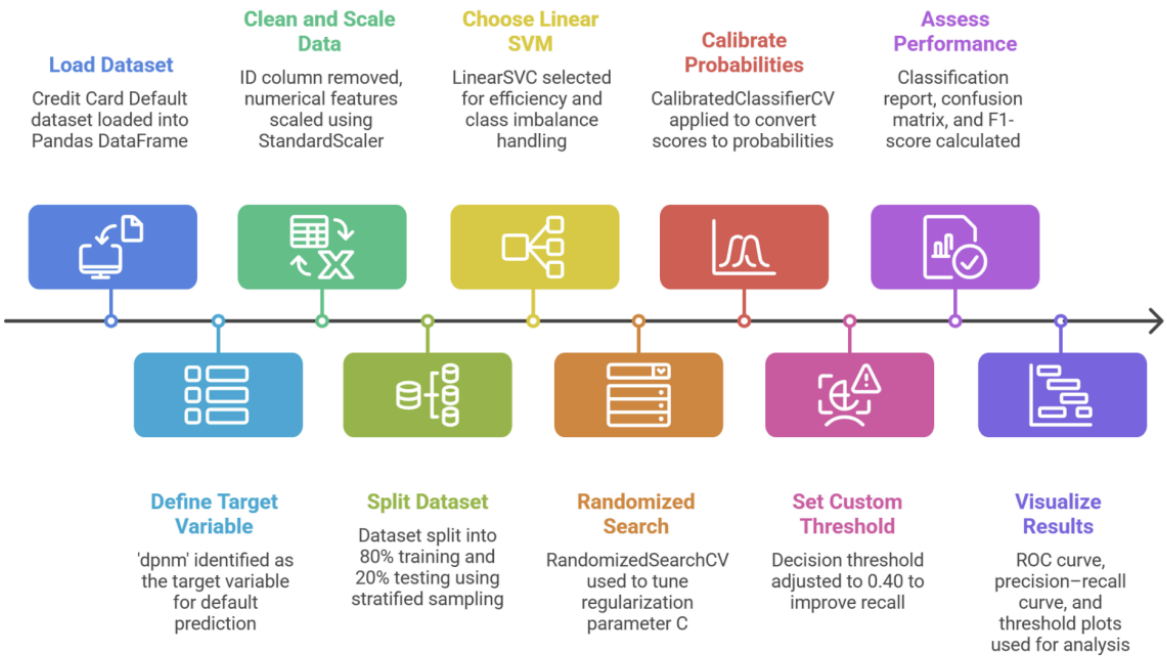
1. Import required libraries
2. Load the Iris dataset
3. Data exploration
4. Feature scaling using StandardScaler
5. Train-test split
6. Train the SVM model with default parameters
7. Perform prediction
8. Apply hyperparameter tuning using GridSearchCV
9. Train an optimised SVM model
10. Compare performance before and after tuning

11. Evaluate the final model

Workflow Diagram:



Dataset → Feature Scaling → Train/Test Split →
Model Training → Hyperparameter Tuning →
Prediction → Evaluation



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6. Performance Analysis

Evaluation Metrics Used:

- Accuracy
- Confusion Matrix
- Classification Report
    - Precision
    - Recall
    - F1-score

Interpretation:

- Higher accuracy indicates better classification
- The confusion matrix shows class-wise performance
- F1-score balances precision and recall

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7. Hyperparameter Tuning

Tuned Parameters:

- C → Regularization parameter
- Kernel → linear / rbf / poly
- Gamma → Controls decision boundary curvature

Tuning Method:

GridSearchCV is used to test multiple parameter combinations and select the best model.

Performance Comparison (Before vs After Tuning)

Before Tuning:

- Model trained with default parameters
- Good accuracy, but not optimal
- May underfit or overfit depending onthe kernel

After Tuning:

- Optimal kernel and parameter values selected
- Improved classification accuracy
- Better generalisation on test data
- More stable decision boundary

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8. Google Colab Link

EXPERIMENT5.ipynb

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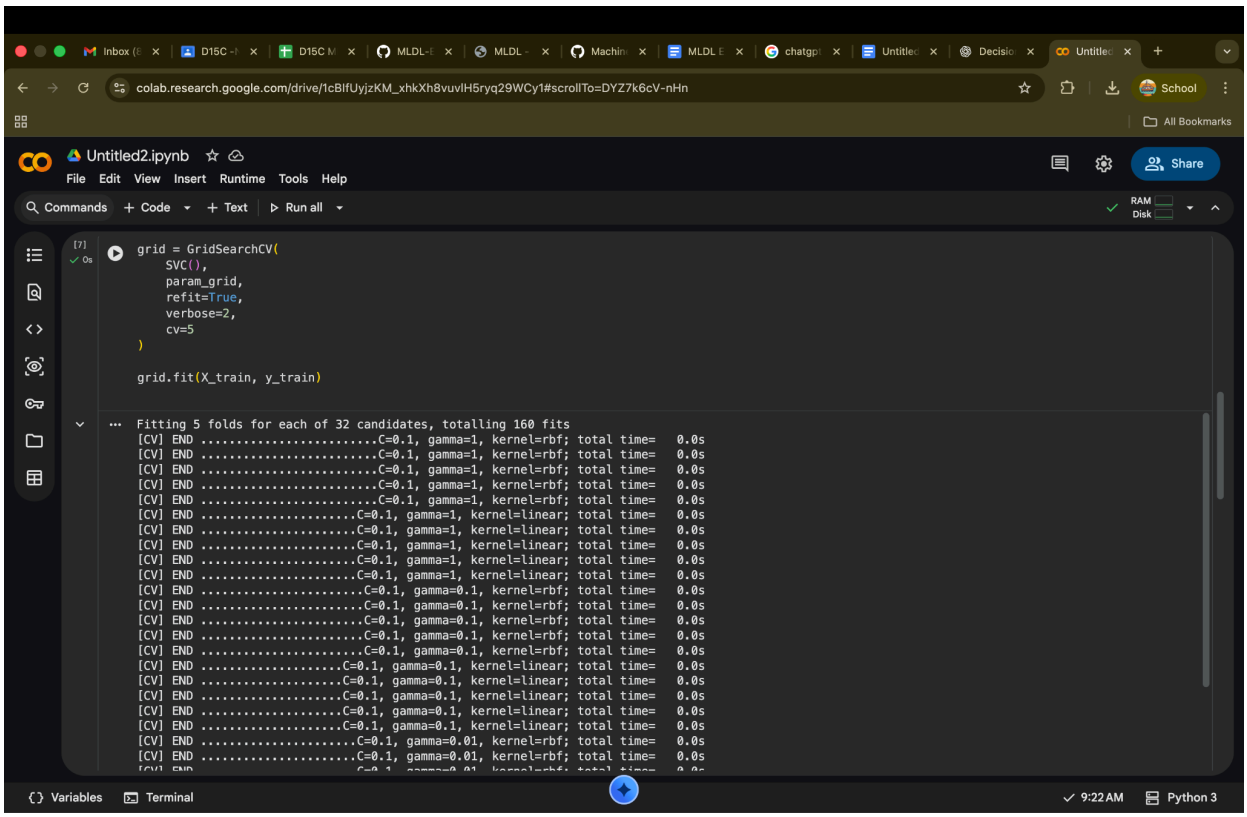
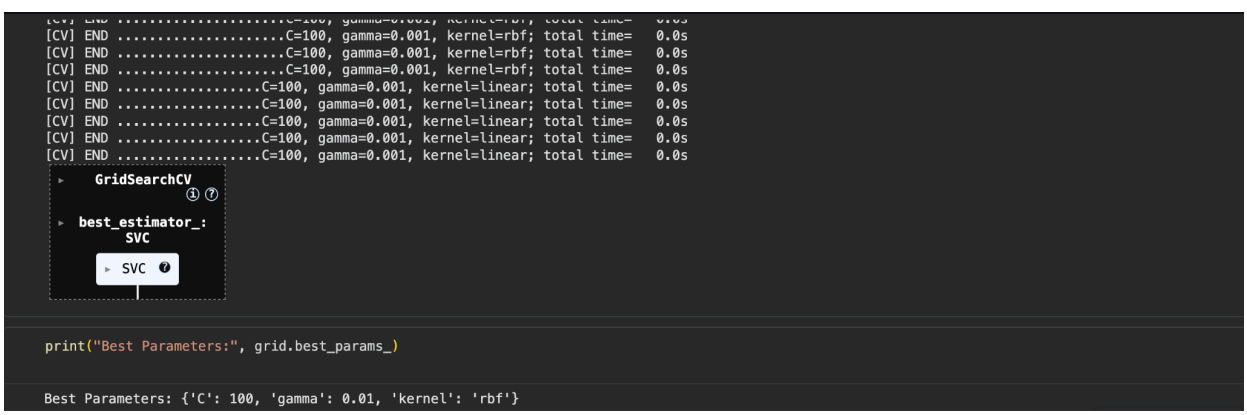
9. Result

The SVM classifier successfully classified the iris flower species.
After hyperparameter tuning, the model showed improved accuracy and better generalisation compared to the default SVM model.





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10. Conclusion

Support Vector Machine was successfully implemented for multiclass classification using the Iris dataset.
Hyperparameter tuning significantly improved model performance by selecting the optimal kernel and regularisation parameters.
This experiment demonstrates the effectiveness of SVM as a powerful margin-based classifier for both linear and non-linear data.